Smoothness-indexed penalized trend estimation in Python: parameter-search coordinates, numerical backends, and reproducible comparisons

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Abstract

Penalized least-squares (PLS) smoothing is established, but its practical computational behavior depends on the difference order, reference-difference mean, numerical solver, optimization coordinate and reuse of intermediate computations. We describe an open-source Python implementation that offers Guerrero's trace-based smoothness through a normalized, order-comparable coordinate $\boldsymbol{S}$ while preserving familiar penalty-based selectors. The article derives the relationship between $\boldsymbol{S}$ and $\boldsymbol{\lambda}$, distinguishes matched-grid search from changes in candidate sets, and evaluates seven numerical backends under an archived 80,640-record Windows factorial design spanning five criteria and two reference means. An independent seed-paired study quantifies full-trend solver speed with bootstrap confidence intervals and Holm-adjusted comparisons; a separate float64 study evaluates four safeguarded inversion algorithms. Archived $CO_{\mathbf{2}}$ and federal funds series illustrate selected smoothness, fitted trends at four difference orders, and historical extrapolation behavior. Results show that both numerical validity and the cost of reusing factorizations and mapped grids matter; no backend is universally fastest, and changing search coordinates does not change the underlying statistical criterion. A reproducible Python API, saved numerical evidence, and render-only graphics complete the implementation contribution.

Keywords: penalized least squares, Whittaker–Henderson smoother, numerical algorithms, smoothness inversion, reproducible benchmarking, time series

1 Introduction

Finite-difference penalized least squares (PLS) provides a flexible way to separate persistent variation from noise. The underlying estimator and its history are well established in the work of Whittaker and Guerrero Whittaker (1923); Guerrero (2007). Cortés-Toto et al. Cortés-Toto et al. (2017) studied attained trend smoothness when traditional CV, GCV, AICc and BIC select the penalty. Efficient structured Whittaker–Henderson computation has an extensive earlier literature Weinert (2007), so neither diagonalization nor smoothness-controlled fitting is presented here as a newly discovered mathematical method.

A modern Python user nevertheless confronts three distinct choices. First, the difference order d and reference difference mean μ determine the geometry of the fitted trend. Second, a statistical selector or a user-specified smoothness target determines the amount of smoothing. Third, the numerical backends and search coordinates determine the computational cost of reaching that same fit. These three issues are often conflated. Our software aims to make them explicit and reproducible.

In particular, a raw penalty λ cannot be compared across difference orders as a universal smoothness scale. We expose a version of Guerrero's trace-based smoothness index normalized by the number of *penalizable* coordinates, so that $S \in [0, 1]$ for all $1 \leq d < n$. This is a monotone rescaling of an established quantity, not a new optimality score. It is, however, a practical interface: a fixed target S can be requested directly, and existing CV/GCV/AICc/BIC or conditional MLE searches can be expressed either in λ or in S. The choice matters for candidate grids, scalar inverse costs, and factorization reuse.

The present contribution is therefore computational and software-centered: (1) an auditable, reusable Python interface that permits controlled smoothness as well as classical criteria; (2) an exact distinction between matched and unmatched parameter grids and their preparation costs; (3) measured end-to-end comparisons over difference orders, fixed means and selection criteria with numerical-validity checks; (4) separate paired uncertainty analysis and numerical-inversion accuracy measurements; and (5) real-data fitted trends and reproducible figures generated from immutable archived results.

For journal-level experimental organization, we draw on the separation between time and quality in Berns et al. Berns et al. (2022), the interpretation of smoothing trade-offs in Gauss et al. Gauss et al. (2021), and explicit algorithmic comparisons in Cardoso et al. Cardoso et al. (2025). These articles concern different algorithms and are presentation comparators, not claims about performance of our PLS software.

2 From penalized least squares to a smoothness coordinate

Let $y = (y_1, \ldots, y_n)^\top$ denote equally spaced observations. For $1 \leq d < n$, define the dth forward-difference operator $D_d \in \mathbb{R}^{(n-d)\times n}$, so $D_1\tau$ contains consecutive increments and $D_2\tau$ contains consecutive differences of increments. The order is not itself a smoothness magnitude: it defines which local changes are penalized. With a reference

difference mean μ, our fitted trend is

$$\widehat{\tau}_{\lambda,\mu} = \arg\min_{\tau}\{\|y-\tau\|_2^2 + \lambda\|D_d\tau - \mu\mathbf{1}\|_2^2\}, \quad \lambda \geq 0. \tag{1}$$

The normal equations give

$$(I + \lambda Q_d)\widehat{\tau}_{\lambda,\mu} = y + \lambda\mu D_d^\top \mathbf{1}, \qquad Q_d = D_d^\top D_d. \tag{2}$$

Since the objective is strictly convex for every finite $\lambda \geq 0$, the solution is unique.

2.1 Null space, order and reference differences

The rows of D_d are linearly independent, and hence $\operatorname{rank}(Q_d) = n - d$. Its null space consists of sequences obtained by evaluating polynomials of degree at most $d-1$ at equally spaced time points; consequently Q_d has exactly d zero eigenvalues. For $\mu = 0$, increasing λ eventually restricts the fitted trend to that polynomial null space. For $\mu \neq 0$ the limiting fitted trend instead satisfies $D_d\tau = \mu\mathbf{1}$; it can contain a nonzero degree-d component. Thus $\mu = 0$ and $\mu \neq 0$ need not yield the same endpoint behavior, particularly when one continues trends beyond the observed sample. The implementation also defines the algebraic edge case $d = 0$ ($D_0 = I$), but the polynomial-null-space interpretation above concerns $d \geq 1$.

Write

$$H_\lambda = (I + \lambda Q_d)^{-1}, \quad b_{\lambda,\mu} = \lambda H_\lambda D_d^\top \mathbf{1}\mu.$$

For a *fixed numerical* μ, the fitted trend is affine in y: $\widehat{\tau} = H_\lambda y + b_{\lambda,\mu}$. Consequently the conditional effective degrees of freedom are $\nu_\lambda = \operatorname{tr}(H_\lambda)$, independent of the fixed reference mean. Estimating μ from the observations changes the full procedure; it does not automatically preserve this conditional EDF.

2.2 Guerrero's index and the order-normalized version

The trace-based smoothness used in the literature of Guerrero Guerrero (2007, 2008) may be written

$$S_G(\lambda) = 1 - \frac{\operatorname{tr}(H_\lambda)}{n}. \tag{3}$$

Because $\lim_{\lambda\to\infty}\operatorname{tr}(H_\lambda) = d$, its maximum for $d \geq 1$ is $(n-d)/n$ rather than one. For comparisons between penalty orders, our package reports the *rescaled* coordinate

$$S(\lambda) = \frac{n}{n-d}S_G(\lambda) = \frac{n - \operatorname{tr}(H_\lambda)}{n-d}. \tag{4}$$

This scaling preserves Guerrero's ordering of smoothing penalties, but puts all orders on the common interval $[0,1]$. It is an adaptation of an established index, *not a new estimator or a novel statistical optimality criterion.* Equal S across orders means the same *fraction of penalizable conditional degrees of freedom*, not the same trend geometry.

inverse mappings: the same S requires different λ across order

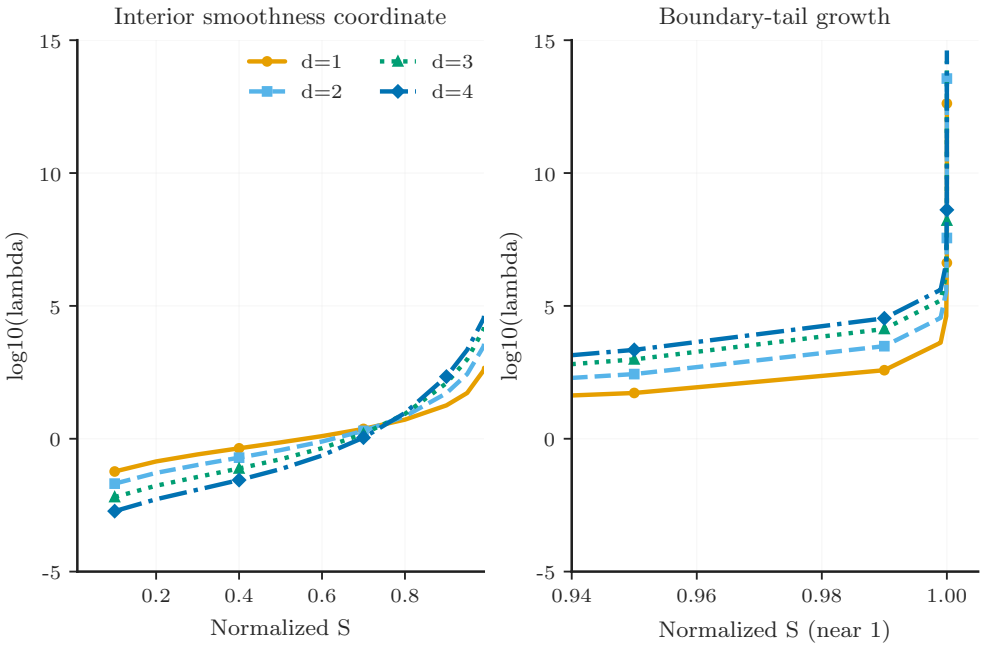


Fig. 1 Archived $S \mapsto \lambda$ inversions for $n = 24$, $d = 1, \ldots, 4$. The left panel covers the interior of the common normalized coordinate; the right enlarges the high-smoothness tail. A common S need not imply a common numerical penalty across difference orders. The exact $S = 1$ boundary has $\lambda = \infty$ and is excluded from logarithmic ordinates.

Let the positive eigenvalues of Q_d be $\delta_1, \ldots, \delta_r$, where $r = n - d$. The d zero eigenvalues yield

$$S(\lambda) = \frac{1}{r} \sum_{j=1}^{r} \frac{\lambda \delta_j}{1 + \lambda \delta_j}, \qquad S'(\lambda) = \frac{1}{r} \sum_{j=1}^{r} \frac{\delta_j}{(1 + \lambda \delta_j)^2} > 0. \tag{5}$$

This proves strict monotonicity for finite $\lambda > 0$, so each $s \in (0, 1)$ has a unique inverse $\Lambda_d(s)$ with $S(\Lambda_d(s)) = s$. The exact limiting values are $\Lambda_d(0) = 0$ and $\Lambda_d(1) = +\infty$. They are handled as analytic boundaries, not by passing infinite penalties to a finite matrix factorization.

2.3 Numerical inversion in log-penalty coordinates

For an interior target s, numerical inversion solves $F(x) = S(e^x) - s = 0$ in $x = \log \lambda$. Differentiation gives

$$F'(x) = \frac{1}{r} \sum_{j} \frac{e^x \delta_j}{(1 + e^x \delta_j)^2}. \tag{6}$$

A bracket ensures that safeguarded Newton, secant, Brent and bisection methods cannot escape the admissible positive-penalty domain. In the tail $s \approx 1$, evaluate

$$1 - S(\lambda) = \frac{1}{r} \sum_j \frac{1}{1 + \lambda \delta_j}$$

directly rather than subtracting two nearly equal floating-point numbers. At $s \approx 0$, likewise use S itself for the residual. A small residual in F is not equivalent to a small relative error in λ at the extreme boundaries; the benchmark records the achieved relative error in the *smaller smoothness tail*. This inversion is one of the explicit costs of offering a human-interpretable smoothness interface.

2.4 Statistical selection is not a change of estimator

For a fixed reference mean, let $C(\lambda; \mu)$ denote one of the already established CV, GCV, AICc, BIC or conditional Gaussian marginal-likelihood scores Cortés-Toto et al. (2017). For example, classical leave-one-out CV can be evaluated through the fixed-μ smoother diagonal:

$$\mathrm{CV}(\lambda) = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{y_i - \widehat{\tau}_i}{1 - h_{ii}} \right)^2, \qquad h_{ii} = (H_\lambda)_{ii}; \tag{7}$$

and GCV is proportional to $n\|y - \widehat{\tau}\|_2^2 / [n - \nu_\lambda]^2$. AICc and BIC use the model's conditional EDF in their information penalties; MLE profiles a Gaussian variance-component likelihood conditional on the specified or plug-in reference difference mean. The criteria have different numerical scales.

Define $\widetilde{C}(s; \mu) = C(\Lambda_d(s); \mu)$. As a mathematical change of variables, global minima agree when the entire domains and boundary conventions are treated identically:

$$\arg\min_{s \in [0,1]} \widetilde{C}(s; \mu) = S\left(\arg\min_{\lambda \in [0,\infty]} C(\lambda; \mu) \right), \tag{8}$$

under the appropriate set-valued interpretation. For an interior differentiable score,

$$\frac{\mathrm{d}\widetilde{C}}{\mathrm{d}s} = \frac{C'(\lambda)}{S'(\lambda)} = \frac{\mathrm{d}C / \mathrm{d}\log\lambda}{\mathrm{d}S / \mathrm{d}\log\lambda}.$$

Stationary points are therefore not created by the coordinate change. However, *finite grids* in s and in $\log\lambda$ generally sample different penalties; a different numerical minimizer can reflect discretization, not a change in the underlying statistical criterion. Our experiments explicitly separate these cases.

Reference-mean caveat.

When fitting, the package permits a fixed numeric μ or the observed-difference plug-in $\widehat{\mu} = \mathrm{mean}(D_d y)$. The public Gaussian MLE selector permits `mu="data"`; classical CV/GCV/AICc/BIC require a fixed numeric μ. For the data-derived comparison, the

numeric estimate is computed from *training data only* and then held fixed inside the classical criterion. This is not the exact CV that would refit μ at every leave-one-out deletion, nor joint MLE of (μ, λ).

3 Reproducible Python implementation

The companion open-source `trendestimation` Python library (version `0.1.0a0` at the evaluated snapshot) is itself a practical outcome: it exposes Guerrero-inspired *smoothness* as a controllable coordinate while retaining the conventional penalty λ for compatibility. We distinguish this software design and numerical implementation from the already known statistical PLS method and classical selectors. We do not claim that this is the first software package for Whittaker–Henderson smoothing.

3.1 Public API and reproducible usage

A reusable `TrendSmoother(n, order=d)` object constructs the difference operator and spectral decomposition once. The user may prescribe a normalized target s, invert it to a penalty and then fit; alternatively, a criterion selects the penalty on either coordinate. The following minimal example shows both workflows and the values returned:

```
from trendestimation import TrendSmoother, select_lambda

smoother = TrendSmoother(len(y), order=2)

# Prescribe normalized smoothness
lam = smoother.lambda_for_smoothness(0.86)
controlled = smoother.fit(y, lambda_=lam, mu=0.0)
print(controlled.smoothness)

# Or select an existing criterion in S coordinates
chosen = select_lambda(
    smoother, y, criterion="gcv", mu=0.0,
    search_space="smoothness",
)
trend = smoother.fit(
    y, lambda_=chosen.lambda_, mu=0.0,
).trend
print(chosen.lambda_, chosen.smoothness)
```

The criterion itself is unchanged; `search_space` chooses the numerical coordinate of its candidate search. The returned `lambda_` can be passed to `fit`, while `smoothness` reports the trace-normalized attained value. The implementation also returns the evaluated penalty grid, scores, and requested coordinate grid for auditing. The established public `TrendSmoother` uses the reusable spectral backend for criterion selection and smoothness inversion. The distinct fit-only convenience `fit_penalized` defaults to a banded Cholesky backend. The new `fit_smoothness` convenience uses spectral inversion internally and does not claim that the banded factorization itself supplies

Table 1 Computational activities and what must be charged. The table is methodological rather than a measured timing ranking. M is the penalty-candidate count and R the number of series. Reused work must be charged once in an end-to-end comparison.

Task	Preprocessing and repeated work	Fair comparison criterion
One fixed trend	Build Q_d or its band; solve for one λ	Fit and setup time, agreement in $\widehat{\tau}$
One scalar criterion	Compute fitted residuals and required EDF/leverage or likelihood diagnostics	Score validity and total evaluation time
Full M-point search	Prepare backend; evaluate M objective scores	Identical candidates, all scores valid
R reused searches	Share spectrum, candidate inversions or factorization	Charge common preparation once
$S^{-1}(s)$ only	Reuse prepared positive eigenvalues; iterate scalar root	Tail error, failure rate and microseconds

the inverse smoothness map. These two public defaults serve different workloads and should not be described as one universal fastest solver.

3.2 Backend differences and caching

The seven numerical implementations evaluated in the large factorial are fresh dense solves, cached dense solves, general banded solves, dense Cholesky, banded Cholesky, symmetric eigendecomposition of Q_d, and full SVD of D_d. These are distinct linear algebra backends, *not* seven statistically different smoothers.

The full SVD expresses the penalty eigenvalues through the squared singular values of D_d. Once y and a fixed μ have been projected, evaluating traces and degrees of freedom for an additional λ costs only a sum over eigenvalues. For a full reconstructed curve, the transformed spectral coefficients must also be mapped back to n coordinates. Therefore a single EDF evaluation, one full fitted trend, an entire CV grid including leverages, and multiple independent searches are different timing problems. A banded $I + \lambda Q_d$ admits sparse-band storage and specialized factorization, often attractive for single fits or larger n Weinert (2007).

3.3 Grid reuse and algorithmic workflow

Figure 2 outlines the computational workflow. Given (n, d), spectral preprocessing depends on the *operator*, not on the particular series. A common grid $s_1, \ldots, s_M$ can be inverted once, cached as $\lambda_1, \ldots, \lambda_M$, and shared among selectors and observed series. That is different from computing a fresh inverse for each criterion evaluation. If the entire grid is used only once, its inversion costs must be included in the end-to-end timing. For a separately spaced log-λ grid, the candidate points differ, so the selected fit may also differ.

3.4 Design for independent reproduction

The repository separates numerical data production, statistical analysis and figure rendering. The 80,640-trial Windows CSV and environment file, the seed-paired solver records, and the scalar root-finding results have been committed independently of all

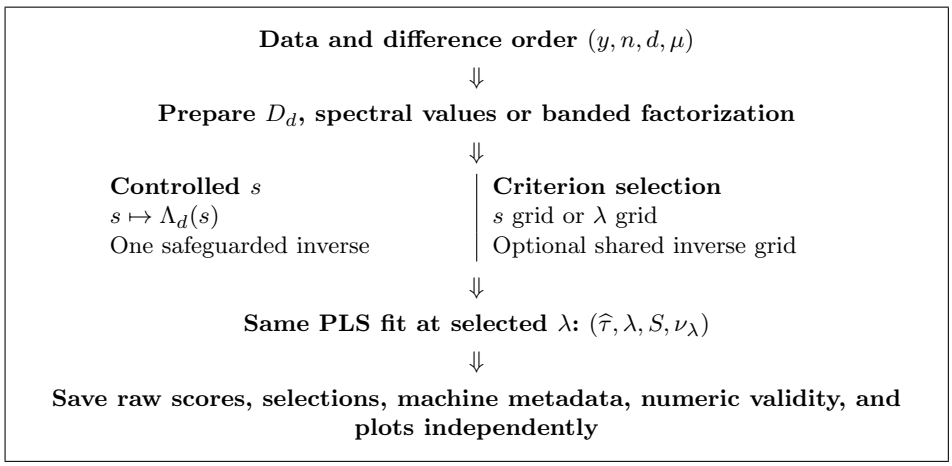


Fig. 2 Reusable software path for controlled smoothness and criterion-based selection. Native-S and once-mapped $S \to \lambda$ grids are equivalent when their candidates match; a separate log-penalty grid generally is not. Each panel of the empirical application reads previously saved selections without repeating benchmark timings.

PDF/SVG/PNG outputs. The plotting scripts verify SHA-256 digests of the numeric archive and reconstruct curves at *stored* (λ, μ). Thus a journal reader may change the color palette or replace a figure layout without rerunning data-generating simulations or hardware-sensitive timing experiments. The independent original draft and complete experiment matrices remain available as supplementary material.

3.5 Computational complexity by workload

Our comparison separates the cost of (i) preprocessing, (ii) one complete trend, (iii) one criterion score, (iv) a search over M candidate penalties, and (v) repeatedly evaluating the same candidates for P criteria or R independent series. These workloads are *not interchangeable*. Let n denote the number of observations and d the fixed difference order. The normal-equation matrix $A_\lambda = I + \lambda D_d^\top D_d$ has half-bandwidth d. Efficient structured Whittaker–Henderson solutions substantially predate our implementation Weinert (2007).

After dense spectral preprocessing, an EDF/trace evaluation and a scalar normalized $S(\lambda)$ evaluation require $O(n)$ operations. Once the observations have been projected, spectral GCV, AICc, BIC and the implemented conditional Gaussian likelihood also admit $O(n)$ score paths per penalty. Full trend reconstruction and the usual spectral diagonal-leverage computation for ordinary CV require $O(n^2)$ per candidate. These costs explain why a fast one-curve banded solver need not be the fastest method for a complete search.

For M distinct *full fitted curves*, two leading cost models are

$$T_{\text{spectral}} = O(n^3 + Mn^2), \qquad T_{\text{banded Cholesky}} = O(Mnd^2).$$

Table 2 Leading operation and memory orders for different numerical solvers. A new penalty entails a new factorization for the direct methods, whereas spectral bases can be reused. CV leverage and trace calculations may add substantial work. The expressions assume $d \ll n$; these are complexities, not measured runtime rankings.

Method	Setup	Full trend per λ	Memory
Dense solve or dense Cholesky	$O(nd^2)$ for Q_d	$O(n^3)$ factorization	$O(n^2)$
General banded solve	Store $2d+1$ diagonals	$O(nd^2)$	$O(nd)$
Banded Cholesky	Store $d+1$ diagonals	$O(nd^2)$ factor; $O(nd)$ solve	$O(nd)$
SVD of D_d	$O(n^3)$	$O(n^2)$	$O(n^2)$
Symmetric `eigh` of Q_d	$O(n^3)$	$O(n^2)$	$O(n^2)$

The spectral cost includes its one-time basis. Banded Cholesky exploits fixed bandwidth but must refactor for each different λ. The constants, BLAS implementation, order and requested diagnostic quantities determine the actual crossover.

3.6 The cost of an S-only coordinate

The normalized mapping $S_d(\lambda)$ depends only on n, d and the positive penalty spectrum; it does not depend on y or on the fixed reference difference mean μ. After preprocessing, one evaluation of S or its derivative costs $O(n)$. A safeguarded inversion with k scalar iterations costs $O(kn)$ per target. A full M-target grid therefore requires $O(Mkn)$ arithmetic *in addition to* the spectral setup. If a banded Cholesky backend does not already have the spectrum, obtaining the mapping has a separate setup cost; the banded factorization is not itself a free S^{-1} routine.

Suppose P criteria use the same M finite S targets. Writing T_{inv} for a candidate inversion and T_{score} for a typical matched criterion evaluation, the two strategies satisfy the idealized accounting identity

$$\begin{aligned} T_{\text{native}} &\approx T_{\text{prep}} + PM(T_{\text{inv}} + T_{\text{score}}), \\ T_{\text{cached}} &\approx T_{\text{prep}} + MT_{\text{inv}} + PMT_{\text{score}}, \\ T_{\text{native}} - T_{\text{cached}} &\approx (P-1)MT_{\text{inv}}. \end{aligned} \tag{9}$$

For unequal criterion costs, replace the common score term by the sum over criteria. This relationship assumes a *matched* candidate set and a one-time cached translation shared among all criteria. At $P = 1$ the conversion cannot generally be eliminated: it is merely moved to an earlier stage. For repeated series with the same (n, d) and target grid, the cached mapping remains reusable.

By contrast, searching in λ and reporting $S(\widehat{\lambda})$ at the end avoids repeated inverse calculations. It can be cheaper for one-off selection. However, sampling a uniform S grid and a log-λ grid produces *different* interior candidates; any different discrete minimum may arise from grid design. Exact monotonic reparameterization preserves the global mathematical minimizer on the full domain.

3.7 Measured solver crossovers and grid reuse

Our paired full-trend results (Table 4) show that no backend universally wins. For $n = 32, d = 2, \mu = 0$, the SVD/banded Cholesky total-time ratio was 1.554 for $M = 1$ (pointwise 95% interval $[1.541, 1.568]$), favoring Cholesky, but 0.419 for $M = 64$ ($[0.417, 0.421]$), favoring SVD. At $n = 96, d = 2, M = 16, \mu = 0$, the ratio was 2.093 ($[2.084, 2.101]$), again favoring banded Cholesky. These host-specific measurements do not justify a universal crossover threshold.

An independent Windows result (Table 3) shows the value of reusing the *same* translated grid across multiple statistical selectors: $n = 48, d = 2, \mu = 0$ required 11.234 ms for five repeated native S searches, compared with 3.980 ms for one S-to-λ translation plus five backend searches. The ratio 2.823 is a descriptive measured advantage of avoiding redundant inversions, not proof that S is intrinsically faster than λ. An independent log-λ grid is not an exact matched comparison.

3.8 A public interface that exposes S, not λ

The package can make normalized S the only *required user-specified* smoothing parameter while retaining λ internally as a numerical representation. The new additive `fit_smoothness` interface calls the existing, validated S^{-1} algorithm and then the usual PLS solver:

```
from trendestimation import fit_smoothness

result = fit_smoothness(
    y, smoothness=0.86, order=2, mu=0.15,
)
trend = result.trend
```

The returned penalty remains available as optional reproducibility metadata. Reusing `TrendSmoother.fit_smoothness` avoids repeating spectral setup between calls, although repeated grid inversions should themselves be cached when appropriate. The familiar λ-based API remains supported for compatibility. Thus eliminating λ from the interface does *not* eliminate its internal computation or constitute a new estimator.

4 Design of the archived numerical studies

We now describe the computational studies in enough detail to distinguish a statistically meaningful result from a mere timing difference. Experiments were performed before figure rendering and their original CSVs are reused verbatim.

4.1 Full-factorial backend and search comparison

The author's Windows full-factorial experiment contains 80,640 recorded search records, not 80,640 statistically independent series. It crosses $n \in \{24, 48\}$, orders $d = 0, \ldots, 5$, fixed reference means $\mu \in \{0, 0.15\}$, five scores (CV, GCV, AICc, BIC and conditional Gaussian MLE), seven backends, and $R \in \{1, 3, 8\}$ independent series

per workload with cold and shared preparation. Two seeds, seven timing repetitions and one BLAS thread were used. The archived environment records Windows, Python 3.11, NumPy 2.4.6, SciPy 1.17.1 and the OpenBLAS version.

The reference grid has seventeen S targets: the deciles between 0 and 1, plus six values concentrated in the near-one tail from 0.95 through $1 - 10^{-12}$. The exact endpoints are analytic; CV/GCV at the $S = 0$ interpolating limit need not admit a finite ordinary value, and Gaussian marginal MLE at $d = 0$ is not applicable under the adopted model. A computational record failing parity with the spectral score is identified as invalid, not allowed to win because its measured runtime happened to be short.

There are five *selection* policies of interest: (i) full-grid native-S search with an inverse at each candidate; (ii) one-time mapped-S penalty grid followed by λ-based search; (iii) a direct finite log-λ grid; (iv) bounded local continuous optimization in S; and (v) bounded local continuous optimization in $\log \lambda$. The first two evaluate *identical* candidates; the third does not, and the two continuous searches use a variable number of objective evaluations. Local continuous minima are *not* evidence of global optimality.

4.2 Independent-seed solver inference

A second author-run Windows study measures preparation plus M complete fitted curves rather than a scalar trace or criterion. It uses 20 independently seeded data sets per predeclared workload and five randomized technical timing blocks per seed. For each paired solver comparison, let $t_{a,i}$ and $t_{b,i}$ be the median full-task times over timing blocks for data seed i. The primary effect is a geometric mean time ratio

$$R_{a/b} = \exp\left[\frac{1}{N}\sum_{i=1}^{N}\log\left(\frac{t_{a,i}}{t_{b,i}}\right)\right].$$

Values above one favor the denominator method. Pointwise 95% percentile bootstrap intervals resample data seeds; two-sided paired sign-flip tests are Holm-adjusted across the predeclared comparison family. The same-host timing blocks and the M fitted penalties are *not* independent inferential units. Sign-flip inference relies on the exchangeability/ symmetry assumption for paired log differences, and all conclusions remain conditional on the recorded hardware and software environment.

4.3 Four root finders in float64

The inversion experiment examines Brent, bracket- safeguarded Newton, bracket-safeguarded secant and bisection with $n = 30, 60, 120$, $d = 0, 1, 2, 3$, eleven targets from 10^{-14} to $1 - 10^{-14}$, and 35 randomized timing blocks per method and target. Every method receives the *same* already prepared smoothness spectrum. Outcomes include successful latency, convergence failure, iterations, function and derivative evaluations, and tail accuracy. The $d = 0$ analytic identity-spectrum inverse is recorded but excluded from numerical-root comparisons. These are technical timing replicates

Table 3 Recorded 17-point Windows complete-search timings (ms), fixed $n = 48, d = 2, R = 1$, spectral SVD, two seeds. The first two policies share all penalty candidates; log-λ has different interior candidates. The final column shows its selected normalized smoothness and is included to make non-equivalence visible. Only numerically valid archive records appear.

Fixed μ	Criterion	S_{native}	Native S	Mapped λ	Log-λ	$S_{\log\lambda}$
0.00	CV	0.850	2.004	1.879	0.545	0.887
0.00	GCV	0.850	1.851	1.800	0.426	0.887
0.00	AICc	0.900	1.887	1.810	0.458	0.887
0.00	BIC	0.100	2.016	1.810	0.449	0.100
0.00	MLE	0.800	3.477	2.035	0.515	0.818
0.15	CV	0.750	1.962	1.869	0.580	0.750
0.15	GCV	0.750	1.933	1.765	0.432	0.750
0.15	AICc	0.800	1.876	1.781	0.439	0.750
0.15	BIC	0.100	1.909	1.791	0.464	0.100
0.15	MLE	0.550	2.033	1.850	0.525	0.458

on one host, not 35 independent data sets; accordingly they do not receive artificial inferential p values.

5 Computational results

5.1 Grid coordinate, method and reference mean

Table 3 reproduces the recorded Windows subset at $n = 48$, $d = 2$, $R = 1$, SVD, comparing every selection criterion and both fixed reference means. The reported unit is milliseconds. The native-S and premapped-S searches select exactly the same S values because the finite candidate set is identical. In contrast, the independently sampled log-penalty grid changes interior candidates and occasionally changes the selected value. Its lower runtime therefore does *not* establish a better way to find the same finite-grid statistical estimator.

For example, in the recorded $\mu = 0$ subset the CV native-S search took 2.004 ms versus 1.879 ms for its pretranslated counterpart; for MLE the corresponding values were 3.477 ms and 2.035 ms. These are *measured*, not modelled complexities. Moreover, when all five criteria shared one translated grid, the recorded sum was 11.234 ms for native-S search versus 3.980 ms for *once-translated plus five backend searches*: a descriptive ratio of 2.823. That accounting charges the common inverse grid only once. Its benefit does not imply that translating a grid is worthwhile for just one inverse target.

The same factorial records distinguish seven numerical backends and all difference orders; numerical failures at extreme finite penalties were screened *before* a timing winner was declared. This is particularly important for higher order penalties: an invalid fast dense solve does not beat a slower accurate spectral score. Full all-cell results, numerical validity flags and reproducibility metadata remain available for inspection in the companion repository.

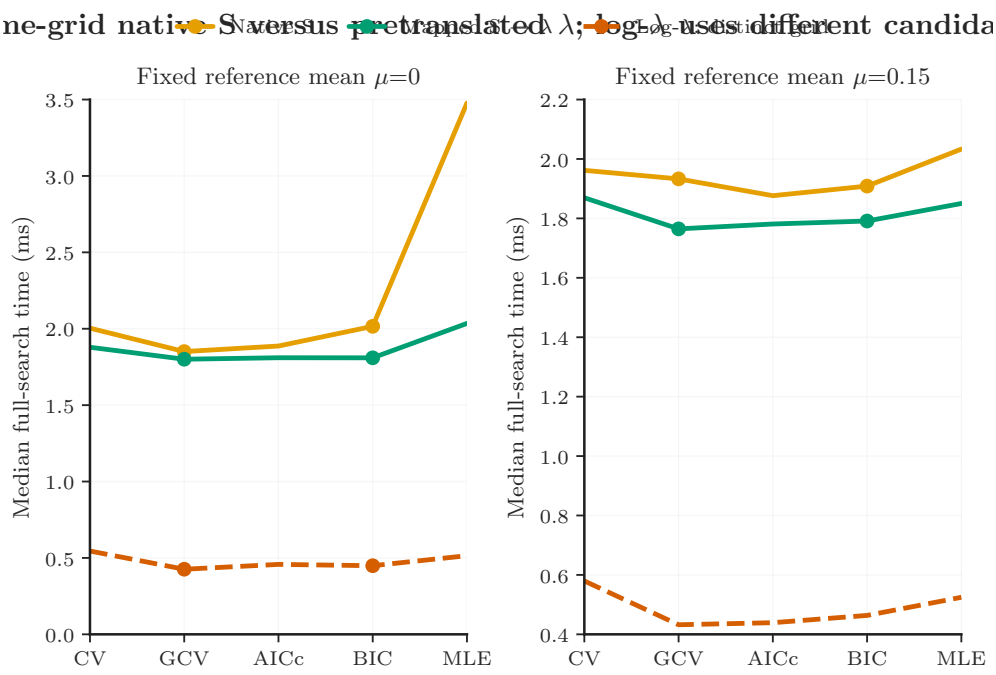


Fig. 3 Recorded end-to-end selection time for each criterion under $\mu = 0$ and $\mu = 0.15$. Native S and premapped $S \to \lambda$ operate on the same seventeen candidates. The distinct log-λ grid is shown as a timing reference rather than a matched-estimator competitor.

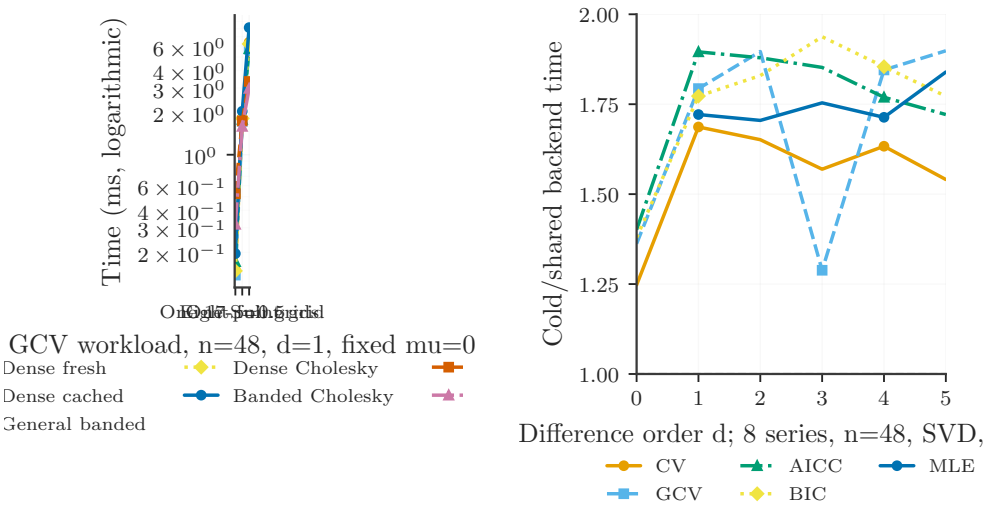


Fig. 4 Previously archived author-Windows 80,640-record study. Left: one scalar score, one complete selection, and repeated searches with backend-specific costs. Right: cold versus shared preprocessing across $d = 0, \ldots, 5$. The backends' validity filters and exact sample settings are provided in the reproducible archived source; the two panels should not be read as cross-hardware generalizations.

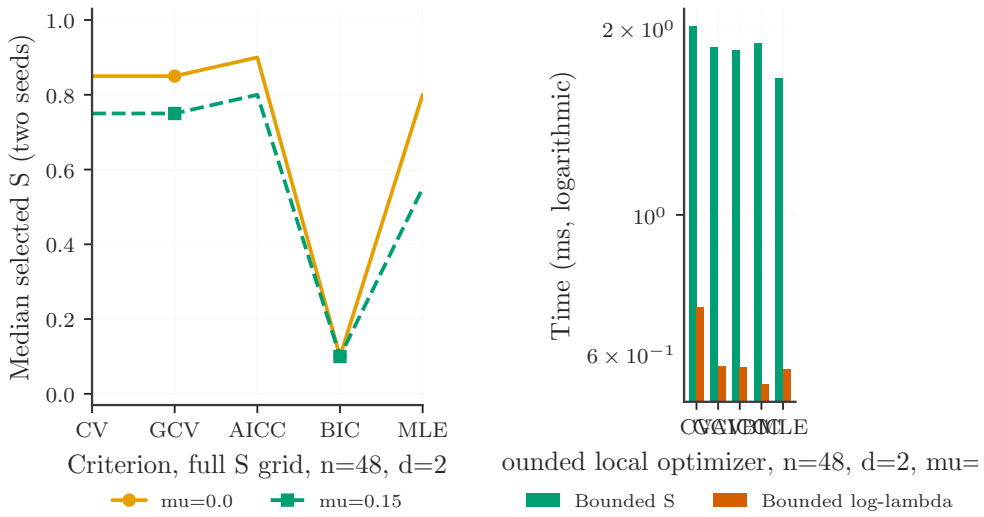


Fig. 5 Archived factorial comparisons: left, selected S across the five statistical criteria under $\mu = 0$ and $\mu = 0.15$ for the matched candidate grid; right, bounded *local* searches over S versus $\log \lambda$. These are separate inferential and computational questions. A bounded optimizer has no guaranteed global minimum in either coordinate.

Table 4 Paired total-time ratio SVD/BChol for archived full-trend computations: ratio > 1 favors banded Cholesky; ratio < 1 favors SVD. The interval is seed-bootstrap pointwise 95%; Holm p covers the complete recorded family of comparisons. M is the number of fitted penalties.

n	d	M	μ	SVD/BChol	95% pointwise CI	Holm p
96	1	16	0.00	2.205	[2.193, 2.216]	0.00315
96	2	16	0.00	2.093	[2.084, 2.101]	0.00315
96	2	16	0.15	2.103	[2.097, 2.109]	0.00315
96	3	16	0.00	1.731	[1.723, 1.742]	0.00315
96	5	16	0.00	1.168	[1.163, 1.174]	0.00315
32	2	1	0.00	1.554	[1.541, 1.568]	0.00315
32	2	64	0.00	0.419	[0.417, 0.421]	0.00315

5.2 Which backend is preferable for full trends?

Table 4 reports seed-paired Windows measurements for distinct difference orders, a nonzero reference mean, and several numbers of fitted curves. These are actual full-trend timings, not the score-only results above. The 95% intervals are *pointwise* bootstrap intervals; Holm-adjusted p values address the full predeclared family of 63 solver-pair/workload comparisons, rather than independent tests chosen after observing the results. Banded Cholesky is a distinct specialized algorithm, not a new name for the older general-banded solve.

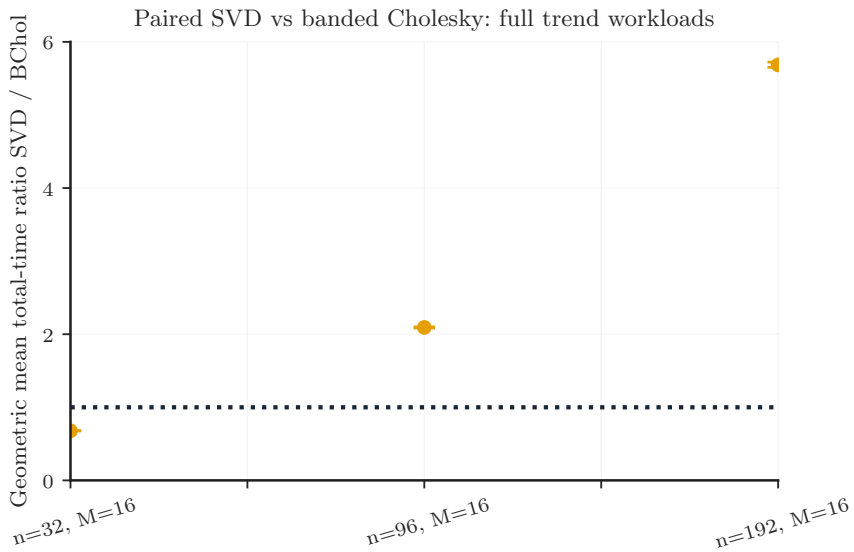


Fig. 6 Archived seed-level paired time ratios for selected full-curve workloads. The dashed horizontal reference of one denotes equal elapsed time, and the intervals are pointwise, not simultaneous. Neither a single backend nor a single configuration is always preferred.

The physical effect of reuse is central: a prepared spectral transform makes repeated (λ, d) evaluations cheap, while a specialized banded factorization can avoid expensive dense preparation for small numbers of full curves. For example, the predeclared $d = 2$ case at $n = 32$ reverses its preferred solver as the number of fitted penalties grows from one to 64. We do not extrapolate this crossover to other numerical precision, sample sizes or hardware.

5.3 Accuracy before inversion speed

The archived inversion results recorded 528 (n, d, s, method) case summaries and no failures across their 35 technical repetitions per case. This statement concerns precisely the tested float64 targets and tolerances; it does not guarantee success for every representable s or penalty spectrum.

The distinction between full search time and one inverse call is important: root-finding is useful for choosing a controlled target but has a cost if repeated at every candidate evaluation. Cache a shared inverse grid where reuse justifies it; do not assume native-S search is faster than a numerical log-penalty search simply because S is easier to interpret.

Table 5 Measured Brent, Newton, secant, and bisection latency, iteration count, failure count and maximum relative smoothness-tail error for $n = 120, d = 2$ at $S = 0.5$ and $S = 0.99$. All values are read from the author's recorded float64 benchmark and require a precomputed smoothness spectrum. The analytic $d = 0$ case is excluded.

S	Inversion method	Median μs	Fail / trials	Iterations	Max rel. tail error
.5	brentq	91.50	0/35	10.0	1.8e-15
.5	newton	88.30	0/35	5.0	2.2e-16
.5	secant	157.70	0/35	24.0	1.1e-15
.5	bisect	292.00	0/35	45.0	1.4e-13
.99	brentq	93.70	0/35	11.0	3.3e-14
.99	newton	118.60	0/35	8.0	0
.99	secant	162.80	0/35	26.0	1.7e-13
.99	bisect	282.00	0/35	45.0	7.7e-14

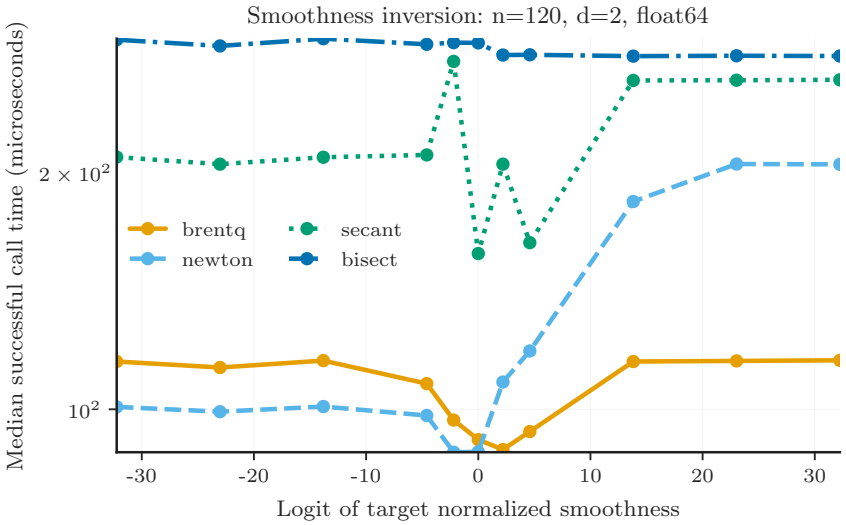


Fig. 7 Observed root inversion latencies across logit(S) for Brent, Newton, secant and bisection. Missing or inaccurate solutions would be explicitly recorded; only valid outcomes contribute successful-call medians. The plotted runtime excludes initial spectral preprocessing.

6 Real time-series illustrations: series, trends and selectors

The final illustration is deliberately separate from the computational timings. It tests how the selected *level* of smoothing and the polynomial order affect an interpretable trend rather than claiming to produce a new stochastic forecast.

6.1 Datasets and common candidate protocol

We use two frozen monthly public series: the Mauna Loa atmospheric CO_2 mean ($n = 441$, January 1990–September 2026, parts per million) and the monthly US

Table 6 Empirical $d = 2$, $\mu = 0$ comparison for the two full series. λ and S are selected in-sample; RMSE and MAE compare a separately extrapolated historical 12-month holdout against raw observations. The 29-point search grid is finite, and the single-origin holdout does not establish predictive superiority.

Series	Criterion	λ	S	RMSE	MAE
CO_2	CV	0.0205	0.100	3.328	2.839
CO_2	GCV	0.0205	0.100	3.328	2.839
CO_2	AICc	0.428	0.509	3.413	2.930
CO_2	BIC	0.0205	0.100	3.328	2.839
CO_2	MLE	0.0205	0.100	3.328	2.839
CO_2	Controlled	41.8	0.860	2.162	1.820
FEDFUNDS	CV	0.0205	0.100	0.512	0.495
FEDFUNDS	GCV	0.0205	0.100	0.512	0.495
FEDFUNDS	AICc	1.21	0.634	0.538	0.522
FEDFUNDS	BIC	0.0205	0.100	0.512	0.495
FEDFUNDS	MLE	0.12	0.320	0.519	0.503
FEDFUNDS	Controlled	39.6	0.860	0.549	0.531

effective federal funds rate ($n = 201$, January 2010–September 2026, percent, not seasonally adjusted) NOAA Global Monitoring Laboratory and Scripps Institution of Oceanography (2026); Board of Governors of the Federal Reserve System (2026). The first contains a strong seasonal component around long-run growth; the second contains abrupt shifts and extended low-rate periods. Both are useful for illustrating how different time scales affect trend estimation.

For each series, the archived study uses $d \in \{1, 2, 3, 4\}$, $\mu = 0$ or a training-derived mean$(D_d y_{\text{train}})$, and CV, GCV, AICc, BIC, conditional Gaussian profile MLE, and controlled $S = 0.86$. The first five optimize their preexisting scores over the *same* 29-point finite S candidate grid; the controlled option selects S by prescription rather than minimization. Consequently a selected $S = 0.1$ or $S = 0.98$ may merely be a *grid boundary* optimum.

6.2 Observed series and four selected trends

Figure 8 plots the *observed monthly values* and fitted trends at the stored $d = 2$, fixed $\mu = 0$ parameters for four contrasting procedures. This view makes visible the substantive consequences of the selection table rather than relying on numerical RMSE alone. Curves are recomputed deterministically at the saved (λ, μ); no statistical criterion is optimized to create the figure.

6.3 Why the difference order matters even at equal S

A simple but important geometric point is that the same normalized trace smoothness does not imply the same fitted curve. At $d = 1$, the limiting zero-reference trend is constant. At $d = 2$, it is linear; larger orders allow correspondingly higher-degree polynomial null spaces. As a result, the finite-λ curves with the same normalized $S = 0.86$ differ as well. Figure 9 plots all four orders for both series, using the same

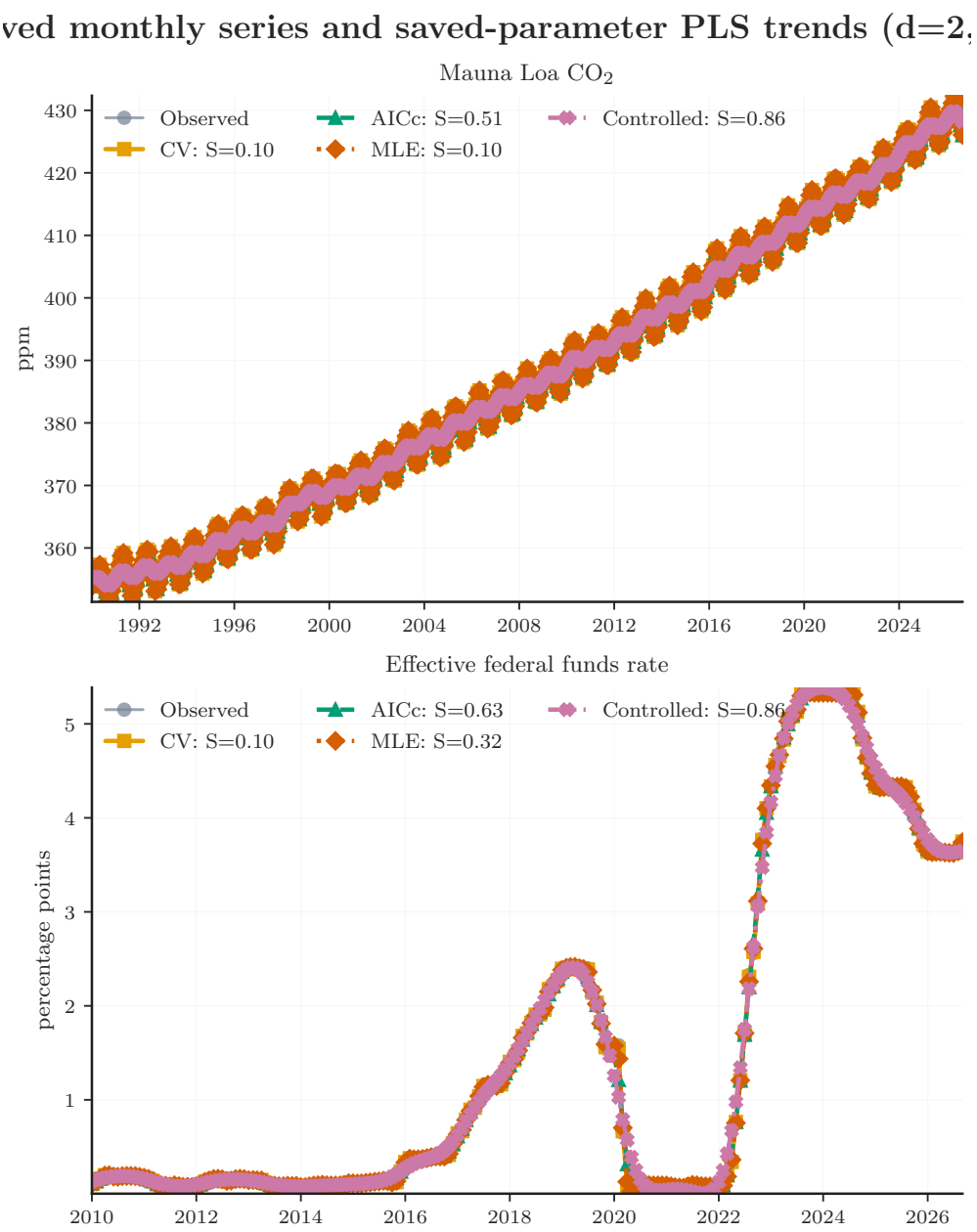


Fig. 8 Frozen monthly observations (gray) and in-sample penalized least-squares trends for CV, AICc, conditional MLE and prescribed $S = 0.86$, with $d = 2$ and fixed $\mu = 0$. The chosen penalty and resulting smoothness level are drawn from saved experiment JSON. Coincident curves may overlap. These are two-sided historical smoothers, not future forecasts.

prescribed smoothness instead of pretending that equal λ values have a comparable meaning.

6.4 Criteria and data-derived reference differences

Figure 10 uses the archived selection and holdout matrices to show two different questions on a common 4×6 layout: how much smoothness is selected, and how much one-origin error is incurred by the *separate* extrapolation rule. The corresponding data-derived μ figure, full 24-panel trend/forecast collections, and the most recent 120-month CO_2 subset are provided as supplementary figures. For μ estimated from observed differences, the numeric value is computed *before* choosing the penalty and separately for each historical training window.

The extrapolation in the archive is an explicit anchored local polynomial fitted to the end of each historical PLS trend, not the PLS estimator itself. For the historical holdout, all mean and penalty selections use the training period only, and the prediction error is measured against *noisy observed values*, not the inaccessible latent trend. For CO_2 the extrapolation has no seasonal model; for the federal funds rate it does not model regime or policy changes. Consequently its one-origin errors are illustrative diagnostics and not the basis for claiming that a particular smoother or root algorithm is a superior forecasting method.

7 Discussion, limitations and conclusion

The distinction among a smoother, its statistical tuning criterion, and the algorithm used to evaluate that criterion is essential. Our normalized S is a trace-based order-adjusted version of Guerrero's index; selecting S directly is a useful software abstraction, but does not create a new estimator. For finite grids, candidate penalties must be identical before timing methods can be interpreted as solving the same statistical selection problem. A separately spaced log-λ grid is a legitimate alternative design, but its chosen fit and selection score may differ.

The archived full-factorial Windows records quantify this distinction across seven backends, five criteria, two fixed means, multiple order choices and computational reuse patterns. The independent seed-paired full-trend analysis supplies pointwise bootstrap intervals and multiplicity-adjusted hypothesis tests on those particular workloads. The inversion benchmark confirms numerical accuracy for the tested targets and reveals costs of offering the more interpretable smoothness-first coordinate. It does not establish hardware-independent superiority, nor should repeated technical timings be treated as independent data seeds. The package's default spectral solver is not claimed to be universally faster than a specialized banded solver.

The real-data case studies illustrate two further limits. The same S need not produce the same geometric curve under different d, and fitting μ from observations changes the interpretation of conditional degrees of freedom. Furthermore, a historical two-sided PLS trend does not define a forecasting process; the archived continuations depend on a separate polynomial extrapolation. Single-origin holdout RMSE does not justify a claim of superior future predictive accuracy. We therefore separate the computational conclusions of this article from forecasting-method development.

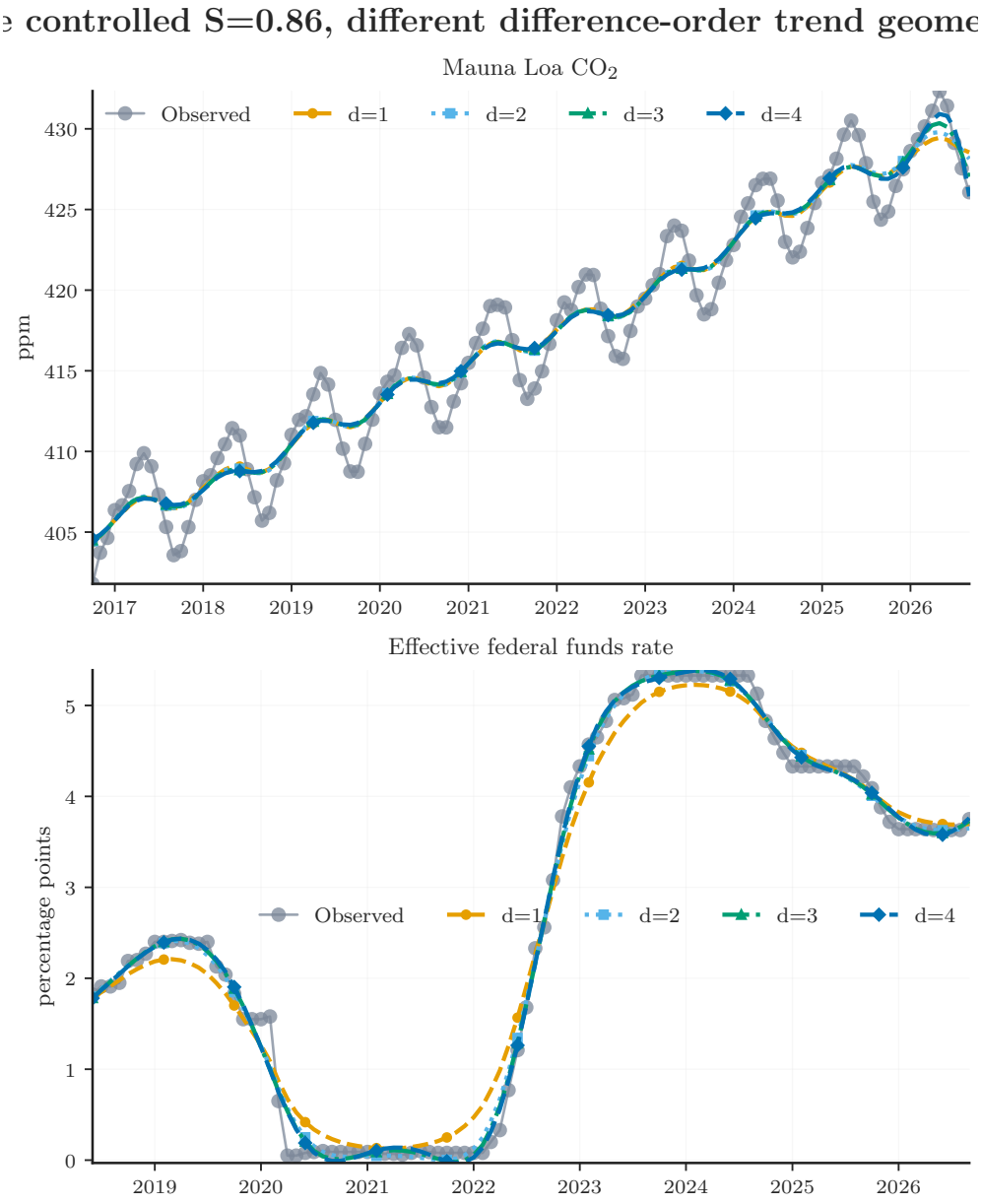


Fig. 9 The most recent monthly observations and four estimated trends at the *same* controlled normalized $S = 0.86$ under $d = 1, 2, 3, 4$, fixed $\mu = 0$. The variation across panels is due to the finite-difference order and its different null-space geometry, not to a change in the prescribed fraction of trace smoothness.

Our main practical result is a reproducible Python implementation of smoothness-indexed PLS that links theory, criterion selection, numerical solvers, and archived evidence. Full high-order matrices, individual candidate scores, complete 4×6 forecasting grids, and machine metadata are provided in the repository rather than crowding the journal text. Future work should investigate larger n, multiple hardware environments, regularized likelihood treatments of unknown μ, and genuinely multi-origin forecasting tests.

Declarations

Funding and competing interests. To be completed and verified by the coauthors before submission.
Data availability. Frozen NOAA and FRED observations, selection-result JSONs, timing CSVs, uncertainty analyses, and rendering manifests are archived in the companion repository. **Code availability.** https://github.com/heri-espino/trendestimation. A permanently archived tagged release should be cited on submission.
Author contributions. To be agreed by the coauthors.

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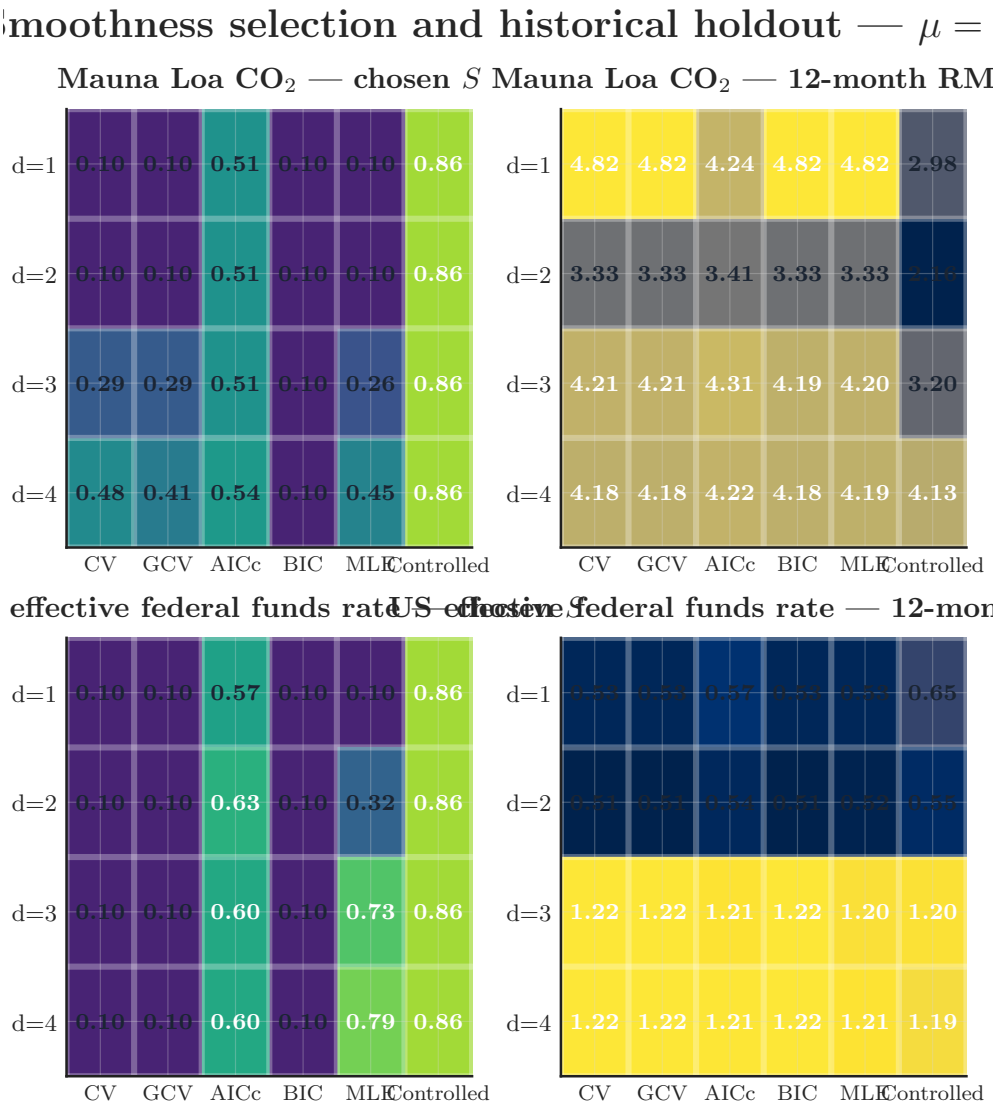


Fig. 10 Saved smoothness selection and historical 12-month holdout errors for the two observed series, four difference orders, and six policies under $\mu = 0$. The six columns include one prescribed smoothness level and five distinct in-sample criterion objectives. Only one historical forecasting origin is represented.